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 Studierichting: Informatica
 Oefeningenreeks 3F nr. 1.9

$$f(x) = \frac{5x^2 - x - 1}{x - 4}$$

VA:

$$\lim_{x \rightarrow 4} \frac{5x^2 - x - 1}{x - 4} = \frac{75}{0} = \infty \Rightarrow \text{VA: } x = 4$$

$$\lim_{x \rightarrow 4+} \frac{5x^2 - x - 1}{x - 4} = \frac{75}{0^+} = +\infty$$

$$\lim_{x \rightarrow 4-} \frac{5x^2 - x - 1}{x - 4} = \frac{75}{0^-} = -\infty$$

HA:

$$\lim_{x \rightarrow \infty} \frac{5x^2 - x - 1}{x - 4} = \lim_{x \rightarrow \infty} \frac{5x^2}{x} = \infty \Rightarrow \text{geen HA}$$

SA:

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{5x^2 - x - 1}{x(x - 4)} = \lim_{x \rightarrow \infty} \frac{5x^2}{x^2} = 5$$

$$\lim_{x \rightarrow \infty} (f(x) - 5x) = \lim_{x \rightarrow \infty} \left(\frac{5x^2 - x - 1}{x - 4} - 5x \right)$$

$$= \lim_{x \rightarrow \infty} \frac{5x^2 - x - 1 - 5x^2 + 20x}{x - 4}$$

$$= \lim_{x \rightarrow \infty} \frac{19x}{x} = 19 \Rightarrow \text{SA: } y = 5x + 19$$

$$\frac{5x^2 - x - 1}{x - 4} - (5x + 19) = \frac{75}{x - 4}$$

$$\frac{x}{x - 4} \left| \begin{array}{c} 4 \\ 75 \\ - \quad | \quad + \end{array} \right.$$

$$\Rightarrow \text{Als } x \rightarrow +\infty \Rightarrow f(x) > A$$

$$\Rightarrow \text{Als } x \rightarrow -\infty \Rightarrow f(x) < A$$

References